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A CRISP-DM Approach to Benchmarking Stochastic,Machine learning and Deep Learning Models in Hydrological Forecasting

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***Abstract*—Flood forecasting in regions with complex tidal and fluvial interactions presents a significant challenge due to non-linear dynamic behaviors. While stochastic, machine learning, and deep learning models have been extensively utilized for water level predictions, isolated evaluations with varying preprocessing pipelines often limit objective performance comparisons. This study addresses this gap by implementing a unified, reproducible pipeline using the CRISP-DM framework to benchmark Stochastic (ARIMA), Machine Learning (Random Forest, XGBoost), and Deep Learning (LSTM) models. Utilizing a rigorously cleaned dataset of 158,828 hydrometric telemetry records from four Irish catchments, the models were evaluated across short-term and long-term forecasting horizons. The results demonstrate that non-linear machine learning models, particularly Random Forest, significantly outperform the linear stochastic baseline, achieving high accuracy for short-term predictions while maintaining stability over extended 24-hour lead times. Furthermore, the integration of an Isolation Forest for anomaly detection and a classification model for extreme event identification enhances the operational reliability of the system. Ultimately, the findings confirm that machine learning and deep learning methodologies provide more robust and viable solutions for real-time flood early-warning systems than traditional stochastic approaches**

Keywords—Flood forecasting , Water level prediction , Machine Learning , Deep Learning , Long Short-Term Memory (LSTM) , ARIMA , CRISP-DM , Hydrological time-series , Random Forest , XGBoost , Anomaly detection

I. INTRODUCTION

Flood forecasting remains a critical challenge in hydrology, particularly in regions where tidal and fluvial interactions introduce complex, non-linear dynamics[8]. Accurate short-term prediction of water levels is essential for early-warning systems, infrastructure protection, and risk mitigation[1]. However, hydrological time-series data exhibit a combination of linear seasonal trends and non-linear stochastic variability, making reliable forecasting difficult[8].

Traditional physical and stochastic models, such as ARIMA, have long been used due to their interpretability and effectiveness in modeling linear dependencies[3]. More recently, Machine Learning (ML) approaches have been adopted for their ability to capture non-linear relationships through engineered features, while Deep Learning (DL) models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in learning temporal dependencies directly from sequential data[5].

Despite these advances, many existing studies evaluate these modeling approaches in isolation, often using different preprocessing pipelines and evaluation strategies. This lack of standardization makes it difficult to objectively compare model performance and limits the practical applicability of findings[3].

To address this gap, this study adopts the CRISP-DM framework to implement a unified and reproducible pipeline for data preparation, feature engineering, and evaluation[10]. By applying this standardized methodology, the study enables a rigorous, side-by-side comparison of stochastic, ML, and DL models on hydrometric telemetry data collected from Irish catchments.

The objective of this research is to benchmark these modeling paradigms to identify the most robust and computationally viable approach for short-term water level prediction in real-time applications.

The study is guided by the following research questions:

- RQ1: To what extent do non-linear ML and DL models outperform linear stochastic models (ARIMA) in catchments influenced by complex tidal patterns?
- RQ2: How does a unified CRISP-DM framework support feature engineering across diverse model architectures?
- RQ3: What are the trade-offs between computational efficiency and predictive accuracy across stochastic, ML, and DL models in a real-time forecasting context?

II. BACKGROUND STUDIES

A. Foundational Data Challenges

Accurate hydrological forecasting is fundamentally dependent on data quality and availability. While physical models provide theoretically grounded predictions, they are often computationally intensive and require extensive parameterization. In practice, hydrometric data collected through sensor networks are frequently affected by sparsity, noise, and missing values due to deployment and maintenance constraints in remote environments [9]. As a result, even advanced modeling techniques may fail when underlying data quality is insufficient.

B. Stochastic Models vs. Machine Learning (ML)

The Stochastic models such as ARIMA have traditionally been used for time-series forecasting due to their simplicity and interpretability. However, their reliance on assumptions of linearity and stationarity limits their effectiveness in highly dynamic hydrological systems.

Machine Learning methods, including Random Forest and Support Vector Machines, have been introduced to address these limitations by capturing non-linear relationships through feature engineering. Nevertheless, comparative studies have shown that ML models do not consistently outperform stochastic approaches, particularly in short-term forecasting scenarios [3]. This observation aligns with the "no free lunch" theorem, which suggests that no single model performs optimally across all problem domains

C. The Dominance and Limits of Deep Learning (LSTM)

To better capture long-term temporal dependencies, Deep Learning models such as LSTM networks have gained prominence. These models are designed to retain historical information through memory cells, enabling improved performance in sequential prediction tasks.

Empirical studies have demonstrated that LSTMs can outperform traditional neural networks in daily streamflow forecasting, particularly for extended lead times [5]. They have also been successfully applied in specialized applications such as peatland water level prediction [6]. Furthermore, optimization techniques such as Bayesian search have been shown to improve model performance and efficiency compared to grid search methods [7].

However, LSTMs require large volumes of high-quality data and may perform poorly when data are sparse or aggregated, such as in monthly forecasting scenarios [5]. Additionally, ensuring reproducibility through fixed random seeds does not necessarily guarantee improved predictive performance [7].

D. Advanced Architectures and Hybridization

Recent research has explored hybrid architectures that combine machine learning with physical principles or multi-scale modeling techniques. For example, dual-scale attention frameworks integrate LSTM and Transformer components to capture both short-term and long-term dependencies, improving predictive accuracy at the cost of increased computational complexity [8].

Similarly, convolutional regression models have been developed to align with physical hydrological concepts such as unit hydrographs, enabling more realistic modeling of flood processes [4]. However, these models can be sensitive to preprocessing choices, particularly interpolation methods for missing data.

In parallel, studies incorporating climate variables and machine learning models such as XGBoost have highlighted the dominant role of precipitation in terrestrial water storage dynamics, while also noting the limitations of excluding anthropogenic factors [2].

Overall, the literature highlights that while advanced models can improve predictive performance, they introduce trade-offs in terms of data requirements, computational cost, and generalizability. These limitations motivate the need for a standardized comparative framework, as implemented in this study.

III. METHODOLOGIES

This study follows the CRISP-DM framework to ensure a structured and reproducible approach to data preparation, feature engineering, modeling, and evaluation. The methodology is designed to enable a fair comparison across stochastic, machine learning, and deep learning models..

A. Data Understanding and Preparations

The primary dataset consists of 172,184 hydrometric telemetry records collected from four Irish hydro station locations: Dublin, Cork, Limerick, and Galway. To preserve temporal integrity, the data were sorted chronologically by station identifier and timestamp.

A structured data cleaning and imputation pipeline was applied:

1. Outlier Removal: Non-physical spikes in water levels were identified using the first-order difference between consecutive observations. Changes exceeding 2.0 meters within one hour were treated as sensor errors and replaced with missing values.
2. Imputation: Missing values were addressed using context-specific strategies. Short gaps (up to 3 hours) in water level and temperature were filled linear interpolation, while battery voltage was forward-filled within a 12-hour limit. Remaining temperature gaps were filled using location-specific mean values.
3. Feature Reduction: Sensor variables with more than 85% missing values were removed. Records with missing target values after preprocessing were excluded, resulting in a final dataset of 158,828 observations

B. Feature Engineering:

To capture temporal and spatial dependencies, several features were engineered:

- Temporal Features: Hour, day of week, and month were extracted to represent cyclical patterns.
- Lag Features: Historical values at previous time steps (t−1, t−2, t−3) were included.
- Rolling Statistics: Rolling mean and standard deviation over a 3-hour window were computed to capture local trends and variability.
- Rate of Change: A velocity feature representing the change in water level between consecutive observations was introduced.

All features were generated within station-specific groups to prevent data leakage across locations.

C. Modeling Architectures

To enable a comprehensive comparison across different modeling paradigms, three classes of predictive models were implemented, each representing a distinct approach to time-series forecasting. All models were evaluated using chronological train-test splits (70/30 and 80/20) to preserve temporal ordering and prevent data leakage.

1. Machine Learning Models

Ensemble-based machine learning methods were selected due to their robustness and ability to capture non-linear relationships without requiring strict assumptions about data distribution.

- **Random Forest Regressor:** Configured with 100 estimators, this model served as a strong baseline due to its ability to reduce variance through aggregation of multiple decision trees and its resilience to noise in the dataset.
- **XGBoost Regressor:** Implemented with 1000 estimators, a learning rate of 0.05, and a maximum depth of 6, this model was used to evaluate the effectiveness of gradient boosting in capturing complex feature interactions and improving predictive accuracy.

2. Deep Learning Model

To capture temporal dependencies directly from sequential data, a Long Short-Term Memory (LSTM) network was developed. Prior to training, all input features and target variables were normalized using Min-Max scaling to ensure stable convergence.

The time-series data were transformed into supervised learning sequences using a sliding window approach, where the previous 6 hours of observations were used to predict the subsequent time step. This approach enables the model to learn short-term temporal dependencies while maintaining computational efficiency.

The network architecture consisted of:

- An initial LSTM layer with 64 units to capture primary temporal patterns
- A Dropout layer (20%) to mitigate overfitting
- A second LSTM layer with 32 units to refine learned representations
- A second Dropout layer (20%)
- Fully connected Dense layers leading to a single output node

3. Stochastic Model (ARIMA)

A traditional linear stochastic baseline was implemented using Auto-Regressive Integrated Moving Average (ARIMA). For each station, the optimal ARIMA order (p, d, q) was automatically selected using the auto_arima algorithm. The model was fitted once on the 80% chronological training set and used to generate forecasts for both short-term (t+1) and long-term (t+24) horizons. This univariate model relies solely on past water level values and serves as the linear stochastic benchmark

The model was trained using the Adam optimizer and Mean Squared Error (MSE) loss function over 10 epochs with a batch size of 64 and a validation split of 10%

D. Horizon Forecasting

To evaluate the practical applicability of the models in real-world forecasting scenarios, performance was assessed across two distinct prediction horizons.

- **Short-term forecasting (t+1):** This scenario focuses on predicting the immediate next time step, which is critical for real-time monitoring systems and rapid response decision-making.
- **Long-term forecasting (t+24):** In this case, the models were tasked with predicting water levels 24 hours in advance. The target variable was generated by shifting the time-series by 24 intervals within each station group. This setup evaluates the ability of models to capture longer-term temporal dependencies and maintain predictive stability over extended lead times.

Assessing both horizons allows for a direct comparison of how predictive accuracy degrades (or remains stable) as lead time increases, which is a key consideration in the design of early-warning systems.

E. Flood Risk Classification and Anomaly Detection

A categorical flood label was defined using the 95th percentile threshold of water levels (3.44 meters). This threshold-based labeling approach enables the identification of extreme events. A Random Forest classifier with 100 estimators was trained to predict the probability of these high-risk conditions.

Anomaly Detection: To distinguish between genuine hydrological extremes and faulty sensor readings, an Isolation Forest algorithm was applied. The model utilized key features including water level, temperature, and battery voltage, and was

configured with a contamination rate of 1% to identify anomalous observations. This step is critical for ensuring data reliability in operational forecasting systems.

F. Evaluation Metrics

Model performance was assessed using task-appropriate evaluation metrics to ensure a comprehensive comparison.

- **Regression Models:** The performance of the Random Forest, XGBoost, and LSTM models was evaluated using:
  - Mean Absolute Error (MAE), to measure average prediction error
  - Root Mean Squared Error (RMSE), to penalize larger errors more heavily
  - Coefficient of Determination (R^2), to assess the proportion of variance explained by the model
- **Classification Model:** The flood prediction model was evaluated using:
  - Precision, Recall, and F1-Score, to assess classification quality
  - Both Macro and Weighted averages, to account for class imbalance

These metrics provide a balanced evaluation of model accuracy, robustness, and practical utility across different predictive tasks.

IV. RESULTS

A. Baseline Model Performance

The performance of the implemented models was evaluated across regression, classification, and anomaly detection tasks, with a particular focus on short-term (t+1) and long-term (t+24) forecasting capabilities. The results provide insight into the comparative strengths of stochastic-inspired ensemble models and deep learning approaches under a unified preprocessing and evaluation framework.

B. Short-Term Forecasting Performance (t+1)

For immediate next-step prediction, the Random Forest Regressor demonstrated strong and consistent performance, achieving a Mean Absolute Error (MAE) of 0.1467 meters, a Root Mean Squared Error (RMSE) of 0.2353 meters, and an R^2 score of 0.9118. These results indicate that the model effectively captures short-term variability and local temporal patterns present in the hydrometric data.

The LSTM model, designed to capture sequential dependencies, achieved a significantly lower MAE of 0.0021 on the scaled dataset. While this result suggests a high degree of precision in learning normalized temporal patterns, it is not directly comparable to the unscaled error metrics reported for the Random Forest model. As such, the LSTM performance should be interpreted as effective within the normalized feature space rather than as a direct improvement in real-world error magnitude.

Overall, both approaches demonstrate strong capability for short-term forecasting, with ensemble methods providing interpretable and stable predictions, and deep learning models showing potential for capturing fine-grained temporal dynamics.

In contrast, the ARIMA stochastic baseline achieved a higher MAE of 0.1894 meters with a negative R^2 of -2.4797, demonstrating that non-linear ML and DL models significantly outperform traditional linear stochastic approaches in tidal-influenced catchments.

C. Long-Term Forecasting Performance (t+24)

To evaluate model robustness over extended prediction horizons, the Random Forest model was applied to a 24-hour forecasting task. The model achieved an MAE of 0.1438 meters, indicating minimal degradation in performance compared to short-term predictions.

This stability suggests that the dataset contains strong temporal structures, such as recurring tidal or seasonal patterns, which can be effectively learned and extrapolated by ensemble-based models. The ability to maintain performance over longer horizons highlights the suitability of such models for early-warning applications where lead time is critical.

The ARIMA model showed further degradation with an MAE of 0.2122 meters and a negative R^2 of -7.1671, further confirming the advantage of ensemble and deep learning methods over linear stochastic models for extended lead times.

D. Flood Risk Classification

The classification framework was evaluated based on its ability to identify high-risk flood events, defined using the 95th percentile threshold of water levels (3.44 meters). The Random Forest classifier achieved a precision of 0.98, recall of 0.96, and an F1-score of 0.97 for the flood class.

These results indicate strong discriminative capability in identifying extreme events within the dataset. However, given the threshold-based labeling and potential class imbalance, these findings should be interpreted with caution, particularly when considering deployment in real-world scenarios where unseen conditions may arise.

E. Anomaly Detection

The Isolation Forest model identified 1,571 anomalous observations within the cleaned dataset of 158,828 records. These anomalies likely correspond to sensor malfunctions, data transmission errors, or rare hydrological events.
The inclusion of anomaly detection provides an additional layer of data validation, ensuring that extreme values used in forecasting are more likely to represent genuine hydrological phenomena rather than measurement errors. This is particularly important for maintaining the reliability of downstream predictive models.

TABLE I. COMPARISON TABLE

Sr.no	Performance			Application
	Models	***metric***	***score***	***domain***
1	XGBoost	R2	0.9317	Regional Forecasting (Dublin)
2	RF Long-term (24h)	R2	0.9118	Long-term Forecasting
3	LSTM (Short-term)	MAE	0.0021	Temporal Deep Learning
4	RF Classifier	F1	0.97	Flood Risk Classification
5	ARIMA	MAE (t+1)	0.1894	Short-term forecasting
6	ARIMA	MAE (t+24)	0.2122	Long-term forecasting

V. CONCLUSION

This study demonstrates that non-linear machine learning and deep learning models significantly outperform traditional linear stochastic models (ARIMA) for water level forecasting in complex, tidal-influenced catchments. By utilizing a standardized CRISP-DM framework, we confirmed that ensemble methods, particularly Random Forest, provide robust and stable predictions across both immediate and 24-hour forecasting horizons. Furthermore, the integration of anomaly detection (Isolation Forest) and extreme-event classification ensures high data reliability. Ultimately, these machine learning approaches offer a computationally viable and highly accurate foundation for real-time flood early-warning systems

While the current models demonstrate strong predictive capabilities, future research should focus on advanced hyperparameter tuning to further maximize efficiency. Specifically, implementing optimization techniques such as Bayesian search could significantly improve model performance compared to standard tuning methods. Additionally, exploring hybrid deep learning architectures that combine LSTMs with Transformer components , or incorporating external meteorological variables like precipitation, could further enhance the model's ability to capture long-term environmental dependencie

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